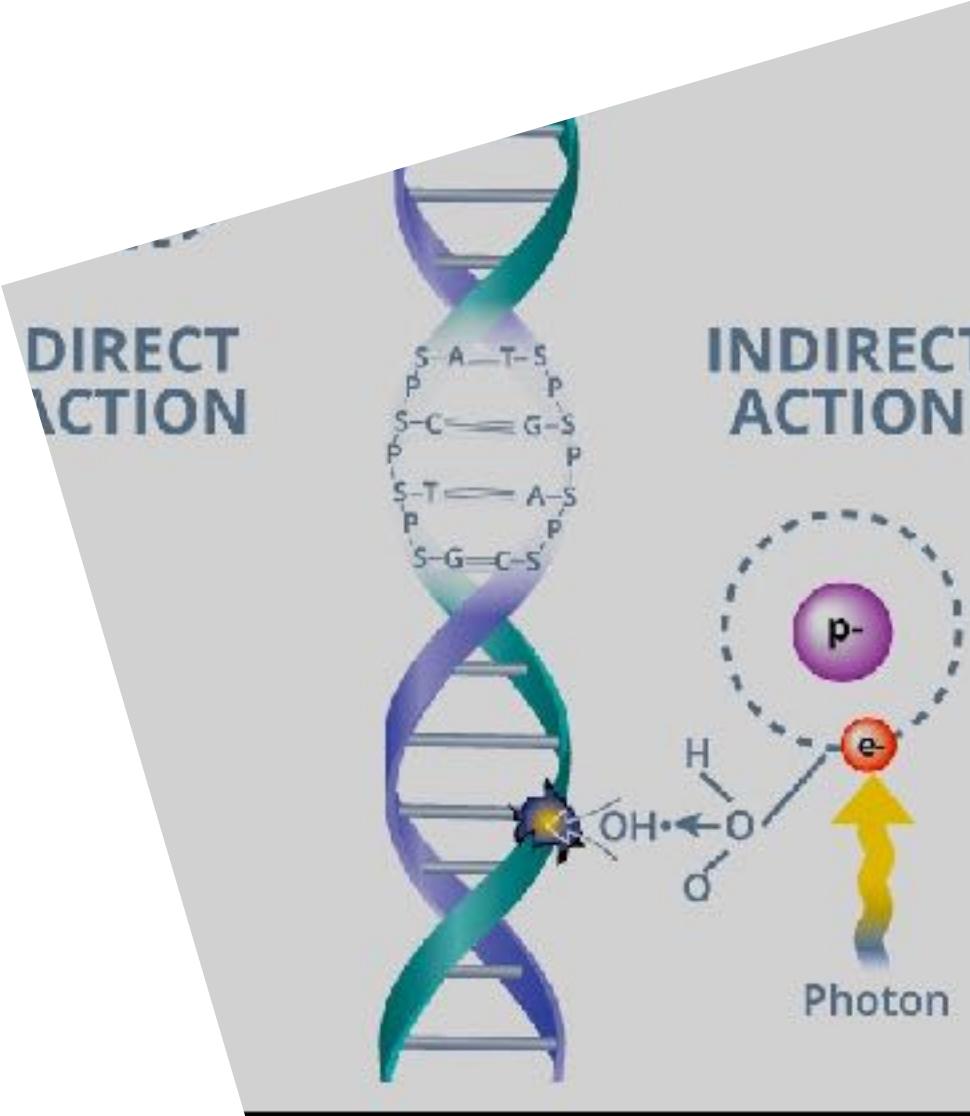
PHYC10009 Physics Foundations

Ionising Radiation and Matter
Week 11 Lecture 2

Professor Phillip Urquijo
School of Physics
University of Melbourne

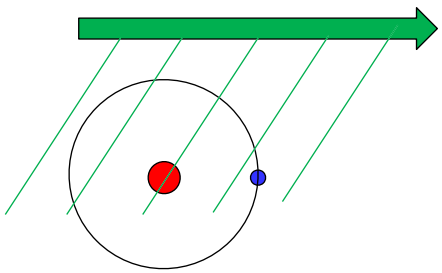


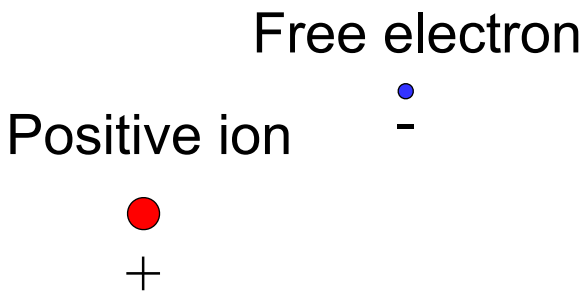
Todays lecture

- Ionising Radiation
- Biological effects of Ionising Radiation
- Radiation dose
- Radiation safety

Ionising radiation G31.4

High energy charged particles like alpha's (α) and beta's (β) ionise atoms as they traverse matter



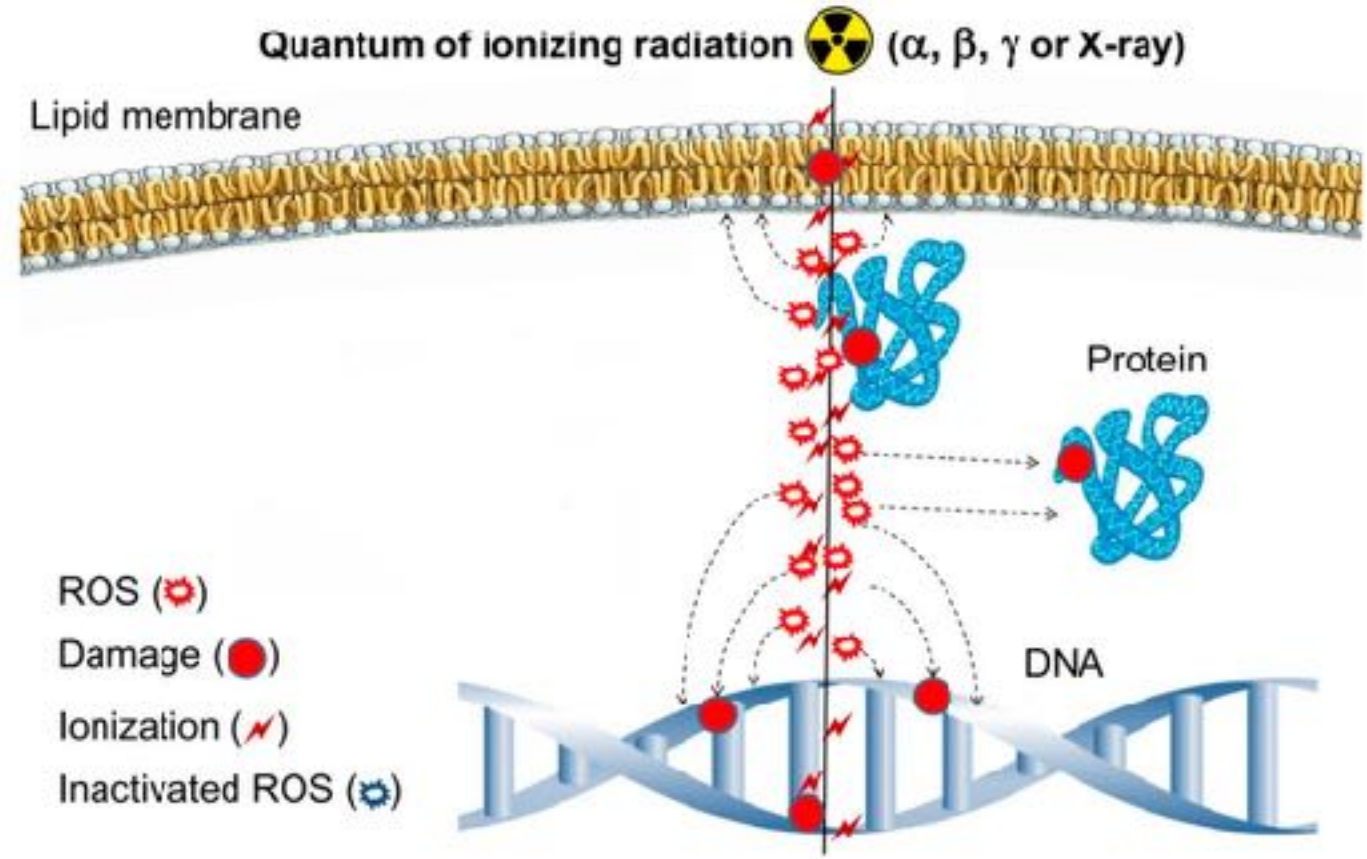


X-rays and gamma rays interact with atoms to eject electrons and create high energy charged particles that then ionise other atoms

A single 1 MeV Alpha particle can ionise 100,000 molecules.

Ionising radiation and Biology G31.4

In biologically active material, which are suspended in water, the effect is to create free radicals $OH^{-}$and H_3O^{+}.
These in turn can damage useful cellular proteins, in particular DNA.



Mars G. Sharapov, Vladimir I. Novoselov and Sergey V. Gudkov https://www.mdpi.com/390270

Biological effects of ionizing radiation G31.4

There are mechanisms in cells to repair damaged proteins.
However, these fail if the damage is too great causing:

(a) Cellular death
(b) Cancerous cells
(c) Genetic mutations if damage is initiated in reproductive systems

Although alpha, beta and gamma radiation can cause cellular damage,
Exposure to these radiation types do not make substances radioactive

Radiation Dose G31.5

The biological effects of ionizing radiation depend on two factors:

The first is how much energy is absorbed by the body.

The second is the type radiation absorbed.

To quantify the first effect, we define a quantity called the absorbed dose.

This is the amount of radiation absorbed by 1 Kg of body tissue.

Dose = (Energy of each radiation quantum)×(Number of quanta/sec)×(time (sec))/(mass)

It's unit is called a Grey, abbreviated to Gy.

1 Gy = 1 Joule/Kg.

So if a sample of 10 Kg of body tissue absorbs 1 Joule of ionising energy, the radiation dose to the tissue is 1/10 = 0.1 Gy.

1 Grey Dose is VERY Dangerous.

Radiation Dose G31.4

We quantify the second effect, type of Radiation with another unit called the "*sievert*" which takes account of Relative Biological Effectiveness (RBE) to make a unit of *dose equivalence* between different types of ionising radiation

The dose equivalent (sievert) = absorbed dose (Gy) x RBE

Radiation type	RBE
X rays	1
Gamma rays	1
Beta particles	1–2
Alpha particles	10–20

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Radiation Dose G31.4

Dose = (Energy of each radiation quantum)×(Number of quanta/sec)×(time (sec))/(mass)

1 Sievert (Sv) is a very big dose and is very dangerous.
More useful are:
micro-Sieverts (μSv) = $10^{-6} Sv$
milli-Sieverts (mSv) = $10^{-3} Sv$

The Yearly Background Radiation from radioactivity in our body, Cosmic Rays, naturally occurring radioactivity and medical activities is 0.5 - 3 milli-sieverts/year
(But this varies a lot with location)

Radiation Exposure

The conservative estimate of the life-time risk due to radiation exposure is that it is the sum of the total exposure.

International Commission on Radiation Protection

- General public < 1 mSv/year *above* background
- Occupational exposure < 20 mSv/year
- Background Radiation is ~ 3.0 mSv/year

Health Risks Arising from Low Doses of Ionizing Radiation

Effect	Risk	Normal Incidence
Risk of cancer from 1 mSv of radiation	1 in 17,000*	57 in 17,000**
Risk of hereditary effect of 1 mSv of radiation	1 in 77,000	1,770 in 77,000

*Age standardised lifetime probability for whole population.
**Age standardised incidence rate for whole population (not necessarily fatal)

The risk of obtaining cancer from 1 mSv of radiation exposure is equivalent to the risk of getting cancer from smoking 400 cigarettes

Sources of radiation

We live in a sea of radiation. Some of it is “natural” background and some is man-made.

1 mSv threshold

Estimated dose (Micro-Sieverts)	Activity
5	sleeping next to your spouse for one year
10	a year of watching TV at an average rate
10	a year of wearing a luminous dial watch
10	a year of living in the USA from nuclear fuel and power plants
10	a day from background radiation (average, varys a lot throughout the world)
20	having a chest x ray
65	flying from Melbourne to London, via Singapore
300	Yearly dose due to body's potassium-40
460	maximum possible onsite dose from Three Mile Island Accident
7,000	having a PET scan
8,000	having a chest CT (CAT) scan
50,000	off-site dose from accident at Chernobyl Nuclear Power Plant (estimates vary widely)
2,000,000	Typical single dose to Cancer region from Radiation Therapy
700,000 - 13,000,000	staff and firefighters at the Chernobyl Nuclear Power Plant during and immediately after the accident
65,000,000	Typical total dose to Cancer region from Radiation Therapy

Typical background exposure is 3000 micro-sieverts per year

Question from 2008 BioMedical Physics exam

The average Human Male body has a mass of 70 Kg and contains 140 grams of Potassium.

a) Given that the ^{40}K isotope contributes 0.012% of the mass of naturally occurring K, calculate the number of ^{40}K nuclei in the average male body.

b) ^{40}K undergoes radioactive decay via beta decay and positron emission. In the following show the number of protons and nucleons for the nuclei that are the products of these decay processes.

c) The half-life of ^{40}K is 1.2×10^9 years. Calculate the number of ^{40}K that decay each second within the human body.

d) On average each nuclear decay of ^{40}K deposits 0.7 MeV of ionising energy within the body. What is the total ionising energy absorbed by the body per year? Calculate your answer in Joules.

e) What is the average dose in <u>milli</u>-Sieverts to the human body per year from ^{40}K (The RBE of the radiation is 1.) ?

Solution to 2008 Exam Question

(a) $\frac{140}{39} \times 6 \times 10^{23} \times \left(0.12 \times 10^{-2}\right) = 2.6 \times 10^{20}$ ^{40}K atoms

(b) (i) e^+ emissions $p \rightarrow n + e^+ + \nu_e$ So A=40, Z=(19-1)=18

(b) (ii) e^- emissions n $\rightarrow p + e^- + \bar{\nu}_e$ So A=40, Z=(19+1)=20

(c) $R = \lambda N_0, \quad \lambda = \frac{1}{\tau}, \; \tau = \frac{t_{\frac{1}{2}}}{\log_e(2)} = \frac{t_{\frac{1}{2}}}{0.693} \Rightarrow \lambda = \frac{0.693}{t_{1/2}}$

$t_{1/2} = 1.2 \times 10^9 \times 3600 \times 24 \times 365.25$ secs

(years) (sec/hours) (hours/day) (days/year)

$t_{1/2} = 3.8 \times 10^{16}$ sec

So $R = \frac{0.693}{3.8 \times 10^{16}} \times 2.6 \times 10^{20} = 4.77 \times 10^3$ sec^{-1}

(d) **Dose = (Energy of each radiation quantum)×(Number of quanta/sec)×(time (sec))/(mass)**

Energy deposited/year = $0.7 \times 1.6 \times 10^{-13} \times 4.7 \times 10^3 \times 3600 \times 24 \times 365.25$

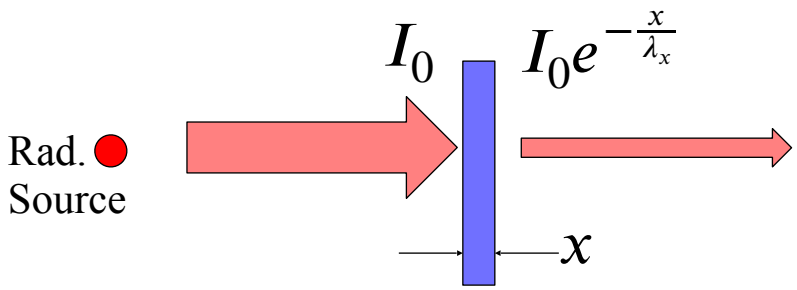
(Energy in J/decay) (decays/sec) (secs/year)

= 0.017 Joules

(e) Dose = $\frac{Energy\ deposited\ in\ J}{Mass\ in\ kg} = \frac{0.017}{70} = 0.24$ mSv

Radiation Absorption – G31.5

	Alpha	Beta	Gamma
Thickness of aluminum to halve intensity	0.0005 cm	0.05 cm	8 cm
Number of ion pairs per mm of air	10^4	10^2	1
Electric charge, Deflection by magnetic fields	+ve, Small deflection	±ve, Large Deflection	0, none
What is it?	$^4He^{++}$	-: electron +: positron	E/M radiation, Short



Radiation Intensity, I as a function of depth, x is:

$$I(x) = I_0 e^{-\frac{x}{\lambda_x}}$$

where λ_x = "Absorption Length"

Radiation Safety G31.5

It is likely you will work with sources of ionising radiation in your career.

Remember, every second of your life you are exposed to ~ 4,700 Bq of radiation from ^{40}K. ie 5,000 Bq is generally safe if external to the body

Ionising radiation dose:

(a) Is limited by the Intensity of the exposure.

(b) Radiation exposure is limited by the length of *time* you are exposed

(Limit the duration and number of exposures.)

(c) Intensity decreases as $\frac{1}{d^2}$ where d = distance to the source

(d) Intensity decreases via the use of material to absorb the radiation $I(x) = I_0 e^{-\frac{x}{\lambda_x}}$

α, β and γ sources are absorbed differently

Radiation Safety G31.5

(a) Is limited by the Intensity of the exposure.

Dose = (Energy of each radiation quantum)×(Number of quanta/sec)×(time (sec))/(mass)

This product of Energy of Quanta × Number of quanta/sec is the intensity

Energy from ^{60}Co γ-ray is 1.25 MeV.

Dose to whole body (70 kg) from 10^4 Bq of absorbed γ-rays for 1 min:

Dose = $1.25 \times 1.6 \times 10^{-13}(J) \times 1 \times 10^4(Bq) \times \frac{60\ (sec)}{70\ (kg)} = 0.0017\mu$Sv

Dose to whole body (70 kg) from 10^6 Bq of absorbed γ-rays for 1 min:

Dose = $1.25 \times 1.6 \times 10^{-13}(J) \times 1 \times 10^6(Bq) \times \frac{60\ (sec)}{70\ (kg)} = 0.17\mu$Sv

Radiation Safety G31.5

(b) Is limited by the time of the exposure.

Dose = (Energy of each radiation quantum)×(Number of quanta/sec)×(time (sec))/(mass)

Time of exposure

Energy from ^{60}Co γ-ray is 1.25 MeV.

Dose to whole body (70 kg) from **10^6** Bq of absorbed γ-rays for 1 min:

Dose = $1.25 \times 1.6 \times 10^{-13}(J) \times 1 \times 10^{6}(Bq) \times \frac{60\ (sec)}{70\ (kg)} = 0.17 \mu$Sv

Dose to whole body (70 kg) from **10^6** Bq of absorbed γ-rays for 1 **hour**:

$$\text{Dose} = 1.25 \times 1.6 \times 10^{-13}(J) \times 1 \times 10^{6}(Bq) \times \frac{3600\ (sec)}{70\ (kg)} = 10\ \mu\text{Sv}$$

(Background Dose ~ 8μSv/day)

Radiation Safety G31.5

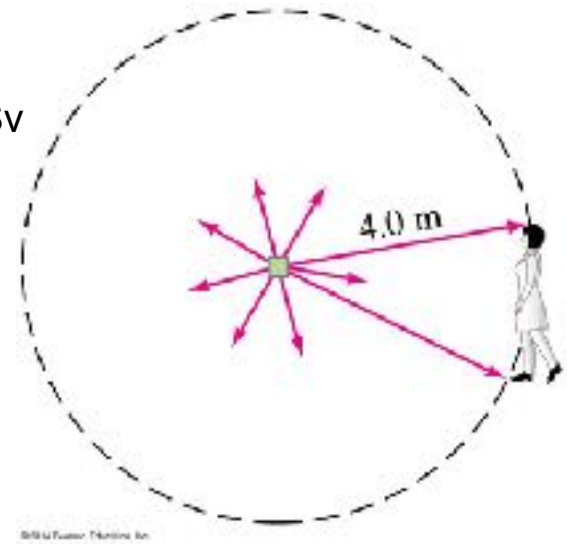
(c) Intensity decreases as $\frac{1}{d^2}$ where d = distance to the source

Energy from ^{60}Co γ-ray is 1.1 MeV.

Intensity at 2 m distance from source is 10^6Bq

Dose to whole body (70 kg) from 10^6 Bq of absorbed γ-rays for 1 hour:

Dose = $1.1 \times 1.6 \times 10^{-13}(J) \times 1 \times 10^6(Bq) \times \frac{3600\ (sec)}{70\ (kg)} = 10\ \mu$Sv



We move twice the distance from the source.

Intensity at 4 m from source = $\frac{1}{2^2} \times 10^6 = \frac{1}{4} \times 10^6 = 2.5 \times 10^5 Bq$

Now dose to whole body (70 kg) from of absorbed γ-rays for 1 hour:

Dose = $1.1 \times 1.6 \times 10^{-13}(J) \times 2.5 \times 10^5(Bq) \times \frac{3600\ (sec)}{70\ (kg)} = 2.5\ \mu$Sv

Radiation Safety G31.5

(d) Intensity decreases with thickness of shielding material.

Energy from ^{60}Co γ-ray is 1.25 MeV.
Intensity at 2 m distance and no shielding from source is 10^6Bq
Dose to whole body (70 kg) from 10^6 Bq of absorbed γ-rays for 1 hour:

Dose = $1.25 \times 1.6 \times 10^{-13}(J) \times 1 \times 10^6(Bq) \times \frac{3600\ (sec)}{70\ (kg)} = 10\ \mu$Sv

Absorption Length of Lead (Pb) is 0.9 cm. We place 3 cm of lead between source and person

Intensity after 3cm of Lead = $10^6 \times e^{-\frac{3}{0.9}} = 10^6 \times 0.036 = 3.6 \times 10^4 Bq$
1 Hour Dose after 3 cm of Lead is:

$$1.25 \times 1.6 \times 10^{-13}(J) \times 3.6 \times 10^4(Bq) \times \frac{3600\ (sec)}{70\ (kg)} = 0.37\ \mu\text{Sv}$$

the absorption length is defined as the distance into a material where the flux has dropped to 1/e of its incident flux

Radiation Dose

Question: A 60 Kg woman takes a 18 hour flight from Melbourne to London. During the flight she is exposed to Cosmic Rays at a rate of 40,000 rays per second. Each Cosmic ray deposits 10 MeV on average. The RBE of Cosmic Rays is 1. What is her total Biological Dose Equivalent from the exposure?

Radiation Dose

Question: A 60 Kg woman takes a 18 hour flight from Melbourne to London. During the flight she is exposed to Cosmic Rays at a rate of 40,000 rays per second. Each Cosmic ray deposits 10 MeV on average. The RBE of Cosmic Rays is 1. What is her total Biological Dose Equivalent from the exposure?

Dose = (Energy of each radiation quantum)×(Number of quanta/sec)×(time (sec))/(mass)

Total Absorbed dose/kg = 4×10^{4}(cosmic rays/sec)x($10\times1.6\times10^{-13}$Joules)x (18x3600 (sec/hour))/(60 Kg) Gy

Absorbed Dose = 6.9×10^{-5} Gy

Biological Dose Equivalent = 6.9×10^{-5} x 1 = 69 μSv

This is about 8 times more than our average daily background dose

3 mSv/year = $\frac{3}{365}$ mSv/day = 8.2 μSv/day

2013 exam

Question 7 **[2 + 5 + 7 + 3 = 17 marks]**

Sarah is a molecular biologist, using ^{32}P as a radioactive label for her experiments. ^{32}P has a 14-day half-life. Sarah uses Perspex (clear acrylic plastic) as a radiation shield. The half-thickness is 6.7 mm; that is, 6.7 mm thick Perspex reduces the radiation intensity by 50%.

(a) Which one or more of the following will halve Sarah's dose rate?

A. Halving the initial quantity of radioactive material
B. Storing the ^{32}P for 7 days before using it
C. Doubling the thickness of Perspex
D. Doubling her typical working distance from the radioactive material
E. None of the above.

(b) ^{32}P decays as follows:

$$^{32}_{15}P \rightarrow ^{32}_{16}S + ? + \bar{\nu}_e$$

(i) What is the missing particle in the above decay equation?

(ii) The missing particle has an average kinetic energy of 694 keV. Calculate the change in mass of a ^{32}P nucleus when it decays, in kilograms.

(c) Given the ^{32}P half-life of 14 days, calculate the decay rate λ and hence the activity in standard SI units, for a sample containing 1.0×10^{15} nuclei of ^{32}P.

2013 Exam Solutions

(a) **A.** Dose = activity (R) ×Energy/particle×time. Since activity is half as much, **Dose drops by a factor two**.

B. Storing for 7 days. $R = 2^{-t/t_{1/2}} R_0 = 2^{-\frac{7}{14}} = 2^{-0.5} = 0.707 R_0$. **This is not a factor of two reduction.**

C. Doubling Perspex shield. Originally:

$R = 2^{-\frac{t}{t_{1/2}}} R_0 = 2^{-1} R_0 = 0.5 R_0$ Doubling t gives:

$R = 2^{-2} R_0 = 0.25 R_0$ **So Dose drops by a factor of two**.

D. Intensity $I(r) = \dfrac{R_0}{r^2}$ so $I(2r) = \dfrac{R_0}{(2r)^2} = \dfrac{R_0}{4r^2}$ **This is not a factor of two reduction.**

E. Clearly not.

(b) (i)

Number of protons increases by one so we have $n \rightarrow p + e^- + \bar{\nu}_e$ so missing particle is an e^-.

(ii) Total energy change ΔE = K.E. + rest mass. Rest mass of electron is 0.511 MeV so:

$$\Delta E = 0.694 + 0.511 = 1.205\ MeV$$

$$\Delta E = 1.205 \times 10^6 \times 1.6 \times 10^{-19} = 1.928 \times 10^{-13} J$$

$$\Delta E = \Delta m c^2 \Rightarrow \Delta m = \frac{\Delta E}{c^2} = \frac{1.928 \times\ 10^{-13}}{3 \times 10^8} = 2.14 \times 10^{-30} kg$$

2013 Exam Solutions continued...

(c) $t_{1/2} = \log_e(2)\tau$ where τ = lifetime and λ the decay rate is $\lambda = \dfrac{1}{\tau}$

$$\Rightarrow \tau = \frac{t_{1/2}}{\log_e(2)} = \frac{14 \times 24 \times 60 \times 60}{\log_e(2)} = \frac{1.21 \times 10^6}{0.693} = 1.75 \times 10^6\ s$$

$$\lambda = \frac{1}{1.75 \times 10^6} = 5.73\backslash\text{times}10^{-7}$$

Activity = $5.73 \times 10^{-7} \times 1.0 \times 10^{15} = 5.73 \times\ 10^8\ Bq$

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Which of the following would reduce the cell damage due to radiation for a lab technician who works with radioactive isotopes:

- Increase the workers distance from the radiation source
- Decrease the time the worker is exposed to the radiation
- Use shielding to reduce the amount of radiation that strikes the worker
- Increase the distance from the source to the worker
- All of the above

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Summary

Dose = (Energy of each radiation quantum)×(Number of quanta/sec)×(time (sec))/(mass)

1 Gy = 1 Joule/Kg.

The dose equivalent (sievert) = absorbed dose (Gy) x RBE

Radiation type	RBE
X rays	1
Gamma rays	1
Beta particles	1–2
Alpha particles	10–20

Ionising radiation dose:

(a) Is limited by the Intensity of the exposure.

(b) Radiation exposure is limited by the length of *time* you are exposed

(Limit the duration and number of exposures.)

(c) Intensity decreases as $\frac{1}{d^2}$ where d = distance to the source

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$$I(x) = I_0 e^{-\frac{x}{\lambda_x}}$$